

**NO OBJECTION CERTIFICATE**

(Required from candidates seeking admission in Ph.D. on Part-Time Basis)

(On the Organization Letter Head)

The undersigned is pleased to permit Mr./Ms. \_\_\_\_\_ who is working in this organization for the last \_\_\_\_\_ years and is presently holding the rank/position of \_\_\_\_\_ for pursuing the Ph.D. programme at Delhi Technological University, Delhi with specialization in the following areas:

1. \_\_\_\_\_ 2. \_\_\_\_\_

To the best of my knowledge and belief Mr./Ms. \_\_\_\_\_ bears a good moral character.

If selected for admission, the candidate will be permitted and be granted leave of absence to be present at the Delhi Technological University as required by the academic schedule to attend classes/ research work. He/She will continue to remain in service of this organization for the entire duration of the course.

This organization has standing commitment to the exemplary standards namely ISO/CMM or similar standard of respective area (required for candidates from industry).

Signature of Head of the  
Institution/Organization with seal

Place: \_\_\_\_\_

Date: \_\_\_\_\_

Name \_\_\_\_\_

Designation \_\_\_\_\_

Email ID: \_\_\_\_\_

## Research Methodology (Common for All)

**UNIT 1-Introduction to Research:** The concept of research, characteristics of good research, Application of Research, Meaning and sources of Research problem, characteristics of good Research problem, Research process, outcomes, application of Research, Meaning and types of Research hypothesis, Importance of Review of Literature, Organizing the Review of Literature. Types of Research : Types of research, pure (basic, fundamental) and applied research, qualitative and quantitative.

**UNIT 2 - Research Design :** Meaning, need, types of research design – Exploratory, Descriptive, Casual research Design, Components of research design, and Features of good Research design. Experiments, surveys and case study Research design. Sampling, Data Collection and analysis: Types and sources of data–Primary and secondary, Methods of collecting data, Concept of sampling and sampling methods – sampling frame, sample, characteristics of good sample, simple random sampling, purposive sampling, convenience sampling, snowball sampling, classification and tabulation of data, graphical representation of data, graphs and charts – Histograms, frequency polygon and frequency curves, bell shaped curve and its properties.

**UNIT 3 - Statistical Methods for Data Analysis:** Applications of Statistics in Research, measures of central tendency and dispersion, Descriptive statistics, Probability functions, Correlation & regression analysis, Statistical inference.

**UNIT 4 - Research Report :** Research report and its structure, journal articles – Components of journal article. Explanation of various components. Structure of an abstract and keywords. Thesis and dissertations: components of thesis and dissertations. Referencing styles and bibliography. Ethics in Research - Plagiarism -Definition, different forms, consequences, unintentional plagiarism, copyright infringement, collaborative work. Qualities of good Researcher.

**UNIT 5 - ICT Tools for Research :** Role of computers in research, maintenance of data using software such as Mendeley, Endnote, Tabulation and graphical presentation of

research data and software tools. Web search: Introduction to Internet, use of Internet and WWW, using search engines and advanced search tools.

## 1. Department of Applied Chemistry

### Discipline: Chemistry

Chemical periodicity, structure and bonding, concepts of acids, bases and non-aqueous solvents, main, transition and inner-transition group elements and their compounds including properties, organo metallic compounds, cages and metal clusters, bio inorganic chemistry.

Basic principles of quantum mechanics, chemical thermodynamics, electro chemistry, chemical kinetics, photochemical reactions, colloids and surfaces, solid state and polymer chemistry.

IUPAC nomenclature of organic molecules, configuration, conformational isomerism in acyclic and cyclic compounds. Organic reaction mechanism, determination of reaction pathways, common named reactions and rearrangements and applications in organic synthesis. Separation, electro- and thermo analytical methods. Molecular spectroscopy and characterization of chemical compounds by IR, Raman, NMR, EPR, UV-vis, MS, electron spectroscopy and microscopic techniques.

### Discipline: Chemical Engineering

Material and energy balance, Chemical Engineering Thermodynamics, Basic concepts of fluid mechanics, flow through pipe, pressure drop calculations, Transport Phenomena, Heat transfer by conduction, convection, radiation, Concepts and design of heat exchangers and evaporators, Basic concepts of mass transfer, Different mass transfer processes and unit operations, Mechanical operations, Chemical reaction engineering: Design of homogeneous and heterogeneous reactors, Chemical process technology, Petroleum refining and petrochemicals. Instrumentation control and optimization, Principles of process economics and cost estimation.

Polymer chemistry, Polymer properties and testing: Determination of molecular weight,